

Cambridge International AS & A Level

CANDIDATE
NAME

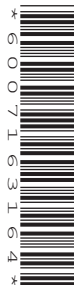
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FURTHER MATHEMATICS

9231/33

Paper 3 Further Mechanics

May/June 2021

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

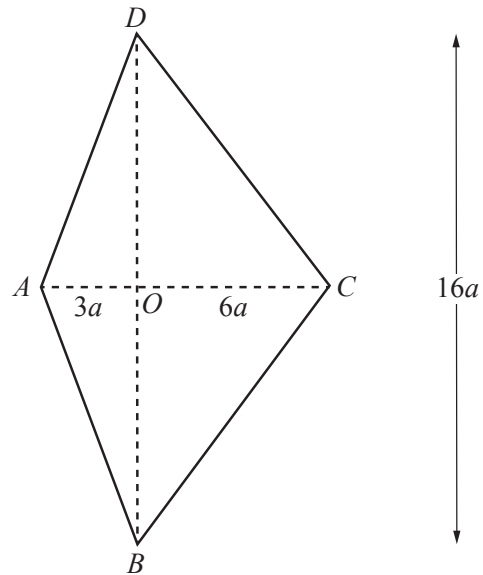
INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 m s^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

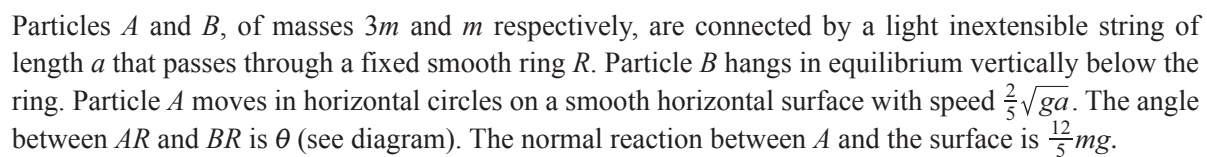


A uniform lamina $ABCD$ consists of two isosceles triangles ABD and BCD . The diagonals of $ABCD$ meet at the point O . The length of AO is $3a$, the length of OC is $6a$ and the length of BD is $16a$ (see diagram).

Find the distance of the centre of mass of the lamina from DB . [3]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

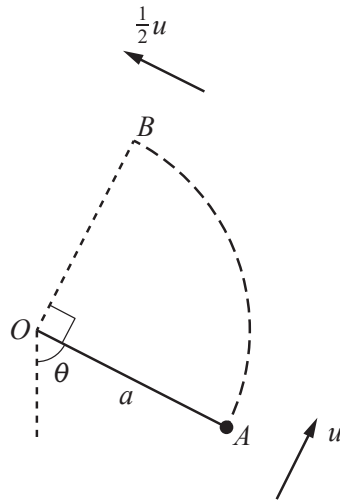
[illegible]



- [illegible]

- (b) Find, in terms of a , the distance of B below the ring. [3]

[illegible]



Find the tension in the string at A , giving your answer in terms of m and g . [8]

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- 5 A particle P of mass m kg is projected vertically upwards from a point O , with speed 20 ms^{-1} , and moves under gravity. There is a resistive force of magnitude $2mv$ N, where $v\text{ ms}^{-1}$ is the speed of P at time t s after projection.

(a) Find an expression for v in terms of t , while P is moving upwards. [6]

[illegible]

The displacement of P from O is x m at time t s.

- (b) Find an expression for x in terms of t , while P is moving upwards. [2]

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- (c) Find, correct to 3 significant figures, the greatest height above O reached by P . [2]

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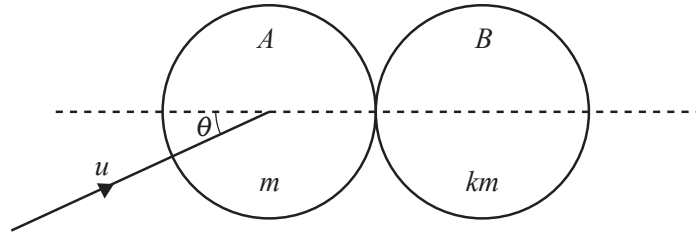
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Two uniform smooth spheres A and B of equal radii have masses m and km respectively. Sphere A is moving with speed u on a smooth horizontal surface when it collides with sphere B which is at rest. Immediately before the collision, A 's direction of motion makes an angle θ with the line of centres (see diagram). The coefficient of restitution between the spheres is $\frac{1}{3}$.

- (a) Show that the speed of B after the collision is $\frac{4u \cos \theta}{3(1+k)}$. [3]

[illegible]

70% of the total kinetic energy of the spheres is lost as a result of the collision.

- (b)** Given that $\tan \theta = \frac{1}{3}$, find the value of k . [6]

[illegible]

- 7 A particle P is projected with speed u at an angle θ above the horizontal from a point O on a horizontal plane and moves freely under gravity. The horizontal and vertical displacements of P from O at a subsequent time t are denoted by x and y respectively.

- (a) Use the equation of the trajectory given in the List of formulae (MF19), together with the condition $y = 0$, to establish an expression for the range R in terms of u , θ and g . [2]

[illegible]

- (b) Deduce an expression for the maximum height H , in terms of u , θ and g . [2]

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

It is given that $R = \frac{4H}{\sqrt{3}}$.

- (c) Show that $\theta = 60^\circ$. [1]

It is given also that $u = \sqrt{40} \text{ m s}^{-1}$.

- (d) Find, by differentiating the equation of the trajectory or otherwise, the set of values of x for which the direction of motion makes an angle of less than 45° with the horizontal. [4]

[illegible]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dashed lines, creating a series of uniform gaps for letter height. The lines are evenly spaced across the entire page, providing a guide for consistent letter formation. There is no text or other markings on the page.

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